Rishi Raj

Doctoral Researcher in Theoretical Physics

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Doctoral researcher in Theoretical and Mathematical Physics at Sorbonne University with a proven record of excelling across multiple fields — from cutting-edge theoretical research and high-performance scientific computing to industry-grade software engineering and machine learning. Skilled in distilling complex problems into elegant computational frameworks, I have designed scalable C++ simulations, built symbolic computation libraries, and engineered production-ready web and cloud systems. My experience spans academic research, open-source contributions, data-driven modeling, and full-stack development, reflecting a rare ability to adapt, learn, and deliver value across diverse domains. With a foundation in problem solving at the highest intellectual level and hands-on expertise in modern software and ML tools, I bring both analytical depth and engineering versatility to tackle challenges in research, technology, and industry.

PROFESSIONAL EXPERIENCE

Doctoral Researcher — Oct 2023 – Present

Supervised by Boris Pioline @ LPTHE, Sorbonne Université — Paris, France

- Extended scattering diagram techniques and BPS spectrum analysis to the local $\mathbb{P}^1 \times \mathbb{P}^1$ Calabi–Yau compactification, pushing the frontier of enumerative geometry in string theory.
- Designed and implemented a high-performance symbolic computation library in Mathematica to compute scattering diagrams to very high orders — released open-source on `GitHub`.
- Derived modular completions of multi-centered BPS indices using localization in supersymmetric quantum mechanics, generalizing prior results beyond the two-center case via generalized error functions.
- Demonstrated exact numerical agreement between quantum indices and string-theoretic modular predictions, unifying insights from supergravity, wall-crossing, and mock modular forms through advanced symbolic and flow-tree-based computation.
- Published two papers (with a third under preparation); presented results at leading conferences and doctoral schools across Europe.

Research Intern — Theoretical Physics — May 2023 – Jul 2023

Supervised by Boris Pioline @ LPTHE, Sorbonne Université — Paris, France

- Selected for the prestigious Charpak Lab Fellowship, hosted jointly by LPTHE and LPENS (Paris); internship extended a prior research stay at LPTHE supported by the IIT Madras Alumni Travel Grant.
- Investigated bound states in $\mathcal{N} = 4$ supersymmetric quiver quantum mechanics (QQM) using symbolic and numerical methods in Mathematica, uncovering structural patterns in the moduli space of multi-centered BPS configurations.
- Conducted a systematic comparison of BPS index computation techniques — attractor flow trees, Coulomb branch localization, and modular anomaly equations — revealing deep connections underlying wall-crossing phenomena.
- Clarified the role of scaling solutions through the Minimum Modification Hypothesis (MMH), achieving exact agreement between attractor tree predictions and localization-based index formulas.
- Leveraged graph-theoretic and high-precision computational techniques to boost performance and accuracy in enumerating physical invariants of black hole bound states.

Web Development Freelancer — Jan 2022 – Jul 2022

Zenoholics Pvt. Ltd. — Bangalore, India

- Contributed to digital transformation solutions deployed across a 300+ client portfolio, driving performance and scalability improvements for high-traffic production systems.
- Led modernization of full-stack web applications using React.js, Next.js, and MongoDB — delivering a 30% performance uplift through architectural refactoring and advanced profiling.
- Built and deployed CI/CD pipelines using GitHub Actions, Docker, and AWS; integrated Kubernetes for scalable, containerized microservice deployment.
- Optimized distributed backend infrastructure with RabbitMQ, connection pooling, caching layers, CDNs, and load testing, enhancing system resilience and responsiveness under scale.

- Diagnosed performance bottlenecks using DevTools, Profiler API, Lighthouse, Postman, and Bundle Analyzer; delivered measurable gains in latency, load time, and throughput.

EDUCATION

Doctor of Philosophy, PhD in Mathematical Physics Oct 2023 – Present
Sorbonne Université — Paris, France
Doctoral Thesis on *BPS Black Holes and Donaldson-Thomas Invariants* under the supervision of Boris Pioline.

Master of Science, MS in Theoretical Physics Aug 2022 – Jul 2023
Indian Institute of Technology, IIT Madras, Chennai, India 9.59/10 GPA
Master's thesis on *Black Holes through the lens of Matrix Theory* under the supervision of Ayan Mukhopadhyay.

Bachelor of Science, BS in Physics July 2018 – Aug 2022
Indian Institute of Technology, IIT Madras, Chennai, India 9.18/10 GPA
Graduated with highest distictions and Minor in Mathematics

ACADEMIC PROJECTS

Free statistics in BFSS matrix model Jul 2025 – Present
Ongoing work with Ayan Mukhopadhyay, Vishnu Jejjala, Tanay Kibe

- Investigating the emergence of free statistical behavior in classical and quantum simulations of the BFSS matrix model by analyzing higher-order free cumulants and spectral data in the large-N limit.
- Applying tools from free probability theory—notably Voiculescu's R-transform and free entropy—to characterize the statistical independence and scrambling dynamics of matrix degrees of freedom.
- Exploring a potential correspondence between Voiculescu's free entropy and black hole entropy, with the aim of uncovering a microscopic basis for thermodynamic behavior in holographic matrix models.
- Informed by recent theoretical developments in quantum chaos, information geometry, and semi-classical gravity, drawing inspiration from Jahnke, Hartnoll-Soni, and Govindarajan et al. to build a bridge between random matrix theory and emergent spacetime.

Improving smuon searches with Neural Networks Feb 2025 – Present
Ongoing work with Mark Goodsell et al.

- Investigating advanced machine learning architectures (**TabNet**, **Particle Flow Networks**, **XGBoost**) to enhance signal-background discrimination in smuon searches at the LHC.
- Working with Monte Carlo simulated datasets to develop robust preprocessing pipelines and feature representations for high-dimensional collider data.
- Exploring adjacent applications such as **detector unfolding** and **systematics-aware training**, aiming to improve interpretability and generalization to experimental data.
- Benchmarking model performance against traditional cut-based analyses to quantify gains in statistical sensitivity and background rejection.

Multi-centered Black Hole Quantum Mechanics and Generalized Error Functions Jan 2025 – Jul 2025
Work with Boris Pioline — see Publications

- Formulated a new saddle-based framework for computing BPS black hole indices in $\mathcal{N}=2$ string compactifications, grounded in the theory of modular completions of Donaldson-Thomas invariants.
- Proved that the refined D4-D2-D0 indices admit an asymptotic expansion governed by modular completions of meromorphic Jacobi forms, including contributions from multicentered black hole configurations.
- Demonstrated deep links between wall-crossing, modularity, and black hole entropy, revealing that discontinuities in DT invariants are smoothly captured by the spectral asymptotics of the modular completions.
- Conducted detailed numerical comparisons between the exact black hole index and its modular approximations, establishing quantitative agreement and validating the proposed physical and mathematical framework.

BPS Dendroscopy on Local $\mathbf{P}^1 \times \mathbf{P}^1$ Mar 2024 – Dec 2024
Work with Boris Pioline and Bruno Le Floch — see Publications

- Constructed scattering diagrams for local F_0 (the canonical bundle over $\mathbb{P}^1 \times \mathbb{P}^1$) across orbifold, large-volume, and Π-stability regimes, enabling a systematic description of the BPS spectrum in type II string compactifications.
- Unified algebraic, geometric, and physical methods (quiver representations, Bridgeland stability, mirror symmetry) to establish explicit wall-crossing structures and extend earlier results from local $\mathbb{P}^2$ to more intricate Calabi–Yau geometries.

- Provided a constructive framework for testing the Split Attractor Flow Tree Conjecture, sketching a proof in this example and clarifying the role of ramification points and mass deformations in stability spaces.
- Developed open-source computational tools (Mathematica package `FOScattering.m`) to reproduce and extend the results, enhancing reproducibility and enabling future research on BPS spectra and enumerative invariants.

Learning Holographic Horizons Jan 2023 – Dec 2023

Work with Ayan Mukhopadhyay, Vishnu Jejjala, Sukrut Mondkar — see Publications

- Pioneered the application of deep neural networks to holography, predicting event and apparent horizon entropies in dynamical AdS black hole spacetimes directly from boundary field theory observables.
- Engineered high-fidelity numerical relativity simulations of asymptotically AdS_4 spacetimes using pseudospectral methods and adaptive time-stepping, generating robust datasets for machine learning analysis.
- Demonstrated universal feature extraction by showing that simple time-series features (maxima, minima, zero crossings of pressure anisotropy) are sufficient to reconstruct horizon dynamics with near-perfect accuracy.
- Opened new pathways for data-driven bulk reconstruction in AdS/CFT, establishing entropy functions as information-theoretic measures that connect boundary one-point functions to emergent spacetime geometry.

Black Holes through the lens of Matrix theory Oct 2022 – April 2023

Master's thesis — unpublished, available on `GitHub`

- Developed and optimized large-scale C++ simulations of the bosonic BFSS matrix model, implementing high-performance numerical solvers that scaled efficiently with matrix size and enabled exploration of chaotic dynamics and thermalization phenomena.
- Engineered rigorous benchmarking pipelines to validate simulations against gauge constraints and conserved charges, ensuring stability and accuracy of long-time evolutions in highly chaotic regimes.
- Performed advanced post-simulation analysis in Mathematica, including spectral comparisons with Gaussian Unitary Ensemble (GUE) statistics, extraction of higher-order cumulants, and identification of deviations from random matrix universality classes.
- Demonstrated emergence of black hole-like microstate behavior in classical matrix dynamics, providing novel evidence for random matrix behavior and free entropy analogues in the context of holographic duality.

PUBLICATIONS

1. Boris Pioline and Rishi Raj (2025). "Multi-centered Black Hole Quantum Mechanics and Generalized Error Functions". In: arXiv: `2507.08551 [hep-th]`
2. Bruno Le Floch, Boris Pioline, and Rishi Raj (2024). "BPS Dendroscopy on Local $\mathbb{P}^1\times\mathbb{P}^1$". In: arXiv: `2412.07680 [hep-th]`
3. Vishnu Jejjala, Sukrut Mondkar, Ayan Mukhopadhyay, and Rishi Raj (2025). "Learning holographic horizons". In: *Phys. Rev. D* 111 (2), p. 026016. DOI: `10.1103/PhysRevD.111.026016`
4. Praveen Pathak, Yogita Patel, and Rishi Raj (2024). "Magnetically Coupled Oscillators and a Smartphone". In: *The Physics Teacher* 62.6, pp. 463–467. DOI: `10.1119/5.0150809`

CONFERENCES AND SUMMER SCHOOLS

4EU+ Ph.D. Summer School on Machine Learning in Physics Jun 2025

@ Niels Bohr Institute, University of Copenhagen — Copenhagen, Denmark

Completed an intensive summer school — covered advanced ML topics including Boosted Decision Trees, Bayesian Neural Networks, Transformers, Anomaly Detection, and Domain Adaptation; trained and optimized a Shakespearean language model; evaluated by an independent committee through lecture participation, hands-on exercises, and model deployment on large structured datasets.

Strings2024 Jun 2024

@ CERN — Geneva, Switzerland

Participated in the premier international conference on string theory, engaging with cutting-edge talks such as Miguel Montero's "String compactifications from top to bottom", Magdalena Larfors' "Machine-learning in Calabi-Yau geometry", and the plenary "100 Open Questions for the Future of String Theory" by Andrew Strominger and Hiroshi Ooguri.

1st ROOT Hackathon Feb 2024

@ CERN — Geneva, Switzerland

Took part in the inaugural ROOT Hackathon, an immersive, open-source development event hosted at CERN's

Idea Square – where contributors of all levels united to tackle existing issues in the ROOT project's issue trackers. I actively contributed by fixing critical bugs, helping enhance the robustness and sustainability of this foundational scientific toolkit.

CERN Winter School on Supergravity, Strings and Gauge Theory Jan 2024
@ CERN — Geneva, Switzerland
Participated in this intensive school organized by CERN's Theory Department, featuring pedagogical lecture series on conformal field theory, cosmology, string compactifications, holography, and black holes. Engaged in discussion sessions and collaborative problem-solving that strengthened my understanding of current frontiers in theoretical physics.

LACES 2023 Dec 2023
@ Galileo Galilei Institute for Theoretical Physics — Florence, Italy
Participated in an intensive PhD-level school designed to bridge Master's coursework and current research in string theory and related fields. The program featured advanced lectures on general relativity, anomalies, supersymmetry, superstring theory, and AdS/CFT correspondence—delivered by leading experts in the field.

Solay Doctoral School on Quantum Field Theory, Strings and Gravity Oct – Nov 2023
@ École Normale Supérieure (ENS) — Paris, France, and CERN — Geneva, Switzerland
Attended this multi-city doctoral school—spread across Amsterdam, Brussels, CERN (Geneva), and Paris—designed to immerse first-year PhD students in advanced theoretical physics. The program featured intensive lecture series led by distinguished faculty, combining rigorous coursework with networking across leading research institutions and fostering international collaboration among emerging researchers.

AWARDS

64th Institute Day — 'Mr S. Venkitaramanan, I.A.S (Retd)' Prize Aug 2023
IIT Madras — Chennai, India
Awarded for outstanding academic and research performance among the graduating batch at IIT Madras.

EDPIF Doctoral Fellowship Jun 2023
Sorbonne University, École Doctorale Physique en Île-de-France (EDPIF) — Paris, France
Highly competitive fellowship supporting doctoral research in theoretical and mathematical physics.

Charpak 2023 Fellow Mar 2023
French Institute in India — Campus France — Paris, France
Prestigious fellowship awarded by the French government to support research internships in France.

Alumni Travel Grant Jun 2022
IIT Madras Alumni Association — Chennai, India
Awarded to support travel and participation in international conferences and research programs.

63th Institute Day — 'Electronics For You' Prize Apr 2022
IIT Madras — Chennai, India
Recognition for excellence in electronics and computing coursework and project performance.

NIUS 2019 Fellow 2019
Homi Bhabha Centre for Science Education (HBCSE–TIFR) — Mumbai, India
Selected as a National Initiative on Undergraduate Science Fellow for training in advanced physics research.

Microsoft Codefundo++ Winner 2018
Microsoft India — Bangalore, India
Secured top position in a national hackathon for developing innovative solutions in disaster management.

KVPY Fellow 2017
Indian Institute of Science (IISc) — Bangalore, India
Recipient of the national Kishore Vaigyanik Protsahan Yojana fellowship for aptitude in research in basic sciences.

TECHNICAL SKILLS

Programming Languages: Proficient in Python (scientific and production-grade), C/C++ (high-performance computing) with CMake, JavaScript / TypeScript (full-stack systems)
ML & AI Frameworks: PyTorch, scikit-learn, XGBoost, Hugging Face, Transformers, Weights & Biases (wandb), Pandas visualization with Seaborn, *R*, ROOT
Cloud & DevOps: End-to-end model deployment pipelines with AWS & GCP, Docker, CI/CD via GitHub Actions, ML model deployment & monitoring, MLOps best practices
Scientific Computing: Mathematica, GNU Scientific Library (GSL), Boost C++, Julia, SageMath

Dev Tools & Version Control: Advanced Git workflows, Linux System Administration, VS Code
Full Stack Web Development: Node.js, Flask, FastAPI, React based frameworks like Next.js, MongoDB, SQL
Core Competencies: Research-grade problem solving, Continuous Learning, Collaborative Problem-Solving, Cross-Functional Communication, Leadership, Adaptability
Langauges: English (native), French (beginner), German (beginner, actively pursuing a langauge course)

OTHER WORK EXPERIENCE

Teaching Assistant Jul 2022 – Nov 2022
PH5060 (Physics Lab I), IIT Madras — Chennai, India

- Assisted students with computational physics problems, including chaotic systems and Monte Carlo simulations.
- Conducted viva sessions and evaluated reports to assess students' understanding and progress.

Horizon Club, IIT Madras Sep 2019
Delivered a lecture series titled *Relativity from Symmetries*, introducing group theory concepts and special relativity to physics enthusiasts

- Designed and presented four lectures focusing on symmetry principles, Lorentz transformations, and physical implications of special relativity.
- Engaged and inspired beginning physics enthusiasts to explore advanced theoretical concepts through interactive discussions.

Music Club, IIT Madras Jul 2019 – Nov 2019
Organized and coordinated musical events, including performances for the annual cultural festival, Saarang

- Led the planning and execution of multiple musical events, ensuring seamless coordination among performers and technical teams.
- Enhanced event management and teamwork skills while handling responsibilities for the institute's cultural programs.